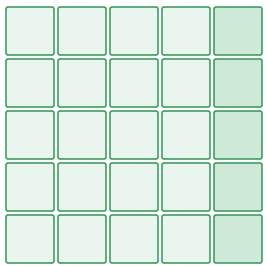
windows3 (one padded pixel in, one window out, II = 1)

padded3 stream
pixel2: 32 ch x 10 b = 320 b

- `line[0]` 259 x 32 ch x 10 b
- `line[1]` 259 x 32 ch x 10 b
- `line[2]` 259 x 32 ch x 10 b
- `line[3]` 259 x 32 ch x 10 b

4 x 82,880 b = 331,520 b -> 36 BRAM18K (ram_2p)

window[5][5][32] 5 x 5 x 32 x 10 b = 8,000 b in registers; shifts one column per pixel



column 4 = new column: 4 rows from line[] + the new pixel
line[] shifts up one row at the same address

windows0: channels 0..15
25 taps x 16 ch x 10 b = 4,000 b, depth 2

windows1: channels 16..31
25 taps x 16 ch x 10 b = 4,000 b, depth 2

The 8,000-bit window is split in two because Vitis HLS rejects aggregated streams wider than 4,096 bits (C09 failed with 'bit-width after aggregation (8000) is larger than 4096').

A window is emitted only for y >= 4 and x >= 4: 65,025 windows from 67,081 padded pixels.

engine3 (continuous_c3_fold, II = 1, 4 cycles per output pixel)

step 0 reads both window halves

step	0	1	2	3
	ic 0..7	ic 8..15	ic 16..23	ic 24..31

step 3 writes 1 output

8 ic x 25 taps = 200 products
`ap_int<10> x ap_int<10> -> ap_int<20>, 200 DSPs (BIND_OP impl=dsp)`

↓

zero-pad to 256, 8-level adder tree
`fabric adders, one register per level -> ap_int<23> partial`

↓

sum = (step 0 ? bias x 2^7 : sum) + partial
`ap_int<23>, fabric add, latency 1`

↓

after step 3: round half to even, saturate
`drop 8 fraction bits -> ap_int<10>, no ReLU`

Pipeline depth 13 cycles (HLS iteration latency): multiply, 8 tree levels, accumulate, quantize.
Weights 32 x 25 x 10 b = 8,000 b in LUTRAM, partitioned 8 x 25 so all 200 taps are read in one cycle.

plain/C17_srcnn_plain.cpp (windows3, engine3); experiments/C17_t4_fifo/reports/engine3_Pipeline_continuous_c3_fold_csynth.rpt; C22 routed: windows3 36 BRAM18K